

EGERTON



UNIVERSITY

UNIVERSITY EXAMINATIONS

REGULAR -NJORO CAMPUS

SECOND SEMESTER, 2023/2024 ACADEMIC YEAR

THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

**(GEOGRAPHY), BACHELOR OF SCIENCE (ENSC), BACHELOR OF ART AND
BACHELOR OF SCIENCE**

GEOG 324: GEOGRAPHIC INFORMATION SYSTEM

STREAM: BSc (Geog), BSc (Ensci), BA & BSc

TIME: 2 HRS

EXAMINATION SESSION: APRIL

YEAR: 2024

INSTRUCTIONS:

- (i) Answer question ONE and any other TWO questions
- (ii) Use sketch maps and diagrams whenever they serve to illustrate an answer
- (iii) Do not write on the question paper
- (iv) Show your working clearly

Question One:

- a) Explain any two data input techniques in GIS **(4 Marks)**
- b) Using examples, explain the following terminologies as used in GIS:
 - i). Positional accuracy **(2.5 Marks)**
 - ii). Conceptual error **(2.5 Marks)**
 - iii). Arc, Vertex and node **(2.5 Marks)**
 - iv). Data model/structure **(2.5 Marks)**
- c) Distinguish between simple feature (spaghetti) and topological vector data models **(5 Marks)**
- d) Explain the main aspects of data quality at micro-level in GIS **(8 Marks)**
- e) Outline any three problems that may arise when paper maps are converted to digital maps **(3 Marks)**

Question Two

Write concise essays on the following:

- a) GIS users (5 Marks)
- b) Advantages of having data in digital format (5 Marks)
- c) Vector model (5 Marks)
- d) Types of errors commonly generated during data input into a GIS (5 Marks)

Question Three

- a) Identify the different data models for the storage of attribute data in GIS (4 Marks)
- b) Discuss why relational DBMS has emerged as the dominant data management tool in GIS implementation and application (5 Marks)
- c) Discuss the practical limitations of scanners as sources of GIS data (10 Marks)

Question Four

- a) Justify the use of GIS in handling geographic data and information (10 Marks)
- b) Discuss the challenges and limitations of GIS implementations and application in developing countries. (10 Marks)

Question Five

- a) Examine the factors considered in the selection of appropriate grid cell size for the raster data model (6 Marks)
- b) Discuss the technical, financial and institutional considerations a GIS implementation plan MUST address (10 Marks)
